IREVOLUTION: ADATA-DRIVEN EXPLORATION OF APPLE'S IPHONE IMPACT IN INDIA USING TABLEAU

TEAM

TEAMID:LTVIP206TMIDS58755

TEAMLEADER:KALARILIKHITH KUMAR

REGISTER NO:SBAP0052594

MAIL-ID : likhithkumarkalari@gmail.com

HALL TICKET NO :233888102021

TEAM MEMBERS:

MATTA MURALI SAI PRAVEEN

MAIL-ID: saipraveenmatta@gmail.com

REGISTER NO: SBAP0052612

HALL TICKET NO : 233888102033

PASUPULETI N.V.R.B.ANIL KUMAR

MAID ID: anilpasupuleti74@gmail.com

REGISTER NO: SBAP0052589

HALL TICKET NO: 233888102037

BOLLA SIVA LOKESH

MAIL ID : lsiva7231@gmail.com

REGISTER NO : SBAP0052587

HALL TICKE NO : 233888102008

INDETI SUNNY

MAIL ID: sunnyindeti@gmail.com

REGISTER NO: SBAP0052628

HALL TICKET NO: 233888102017

IRevolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau

Project Description:

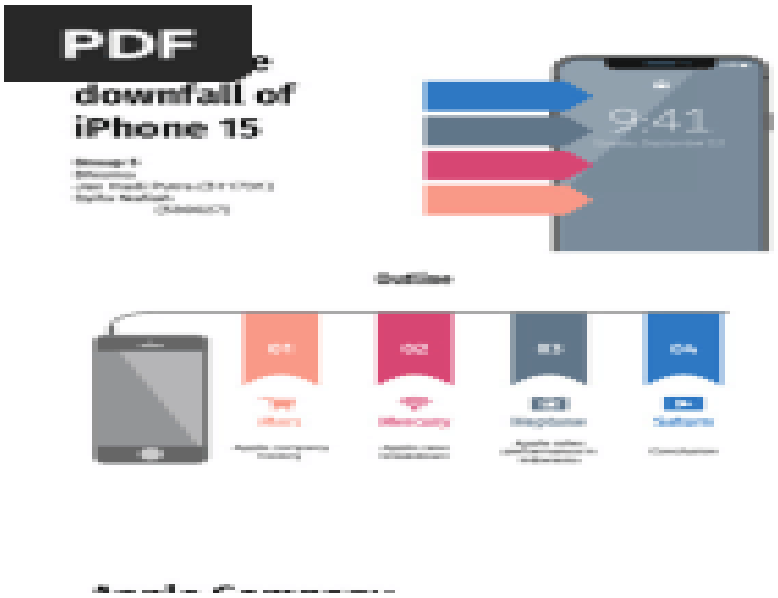
"iRevolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau" is a project that aims to investigate and visualize the influence and effects of Apple's iPhone on the Indian market. Utilizing Tableau's powerful data visualization capabilities, the project explores various aspects such as market penetration, sales trends, user demographics, and the cultural impact of iPhone adoption in India. By examining data from sources like sales records, social media sentiment, and market research, the project provides valuable insights for industry stakeholders, including Apple, local competitors, and market analysts.

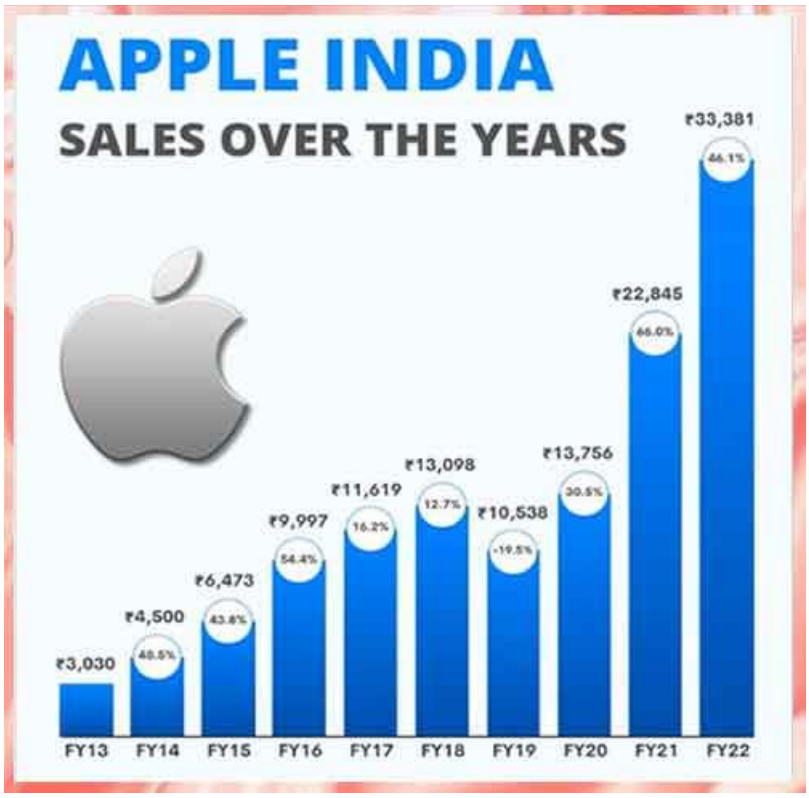
"iRevolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau" is an analytical project that uses interactive, dark-mode dashboards to visualize iPhone's market performance, sales trends, and user preferences in India. It provides actionable insights into regional, price-wise, and feature-wise data, highlighting Apple's rapid expansion in the Indian market to stakeholders.

Key Aspects of the iRevolution Project:

- **Purpose:** The project aims to provide a centralized, story-driven Tableau dashboard to analyze Apple's growing impact in India,, addressing the need for better data-driven decision-making.
- **Key Themes:** Data points include iPhone's market share growth in India, which has doubled to 8% since 2022.
- **Visualizations:** The dashboard typically includes visual insights into:
  - **Market Trends:** Quarterly market share, sales trends, and annual sales, with analysis of over 1.9 billion iPhones sold.
  - **Price & Features:** Analysis of price-wise trends and specific, feature-wise comparisons of various iPhone models.
  - **Regional Impact:** Insights into how the brand is performing across different regions in India.
- **Tools & Design:** The project utilizes Tableau's visualization capabilities to create a "Dark UI" for enhanced user experience and better, more comfortable data interpretation.
- **Contextual Data:** The analysis includes context on Apple's shift in manufacturing, with India now representing a significant portion of global production (25% by 2026), according to this Reuters article.

This project serves as a tool to visualize the shift in India's role from a consumer to a key manufacturing hub, as detailed in reports regarding Tata Electronics' role in the Apple supply chain.



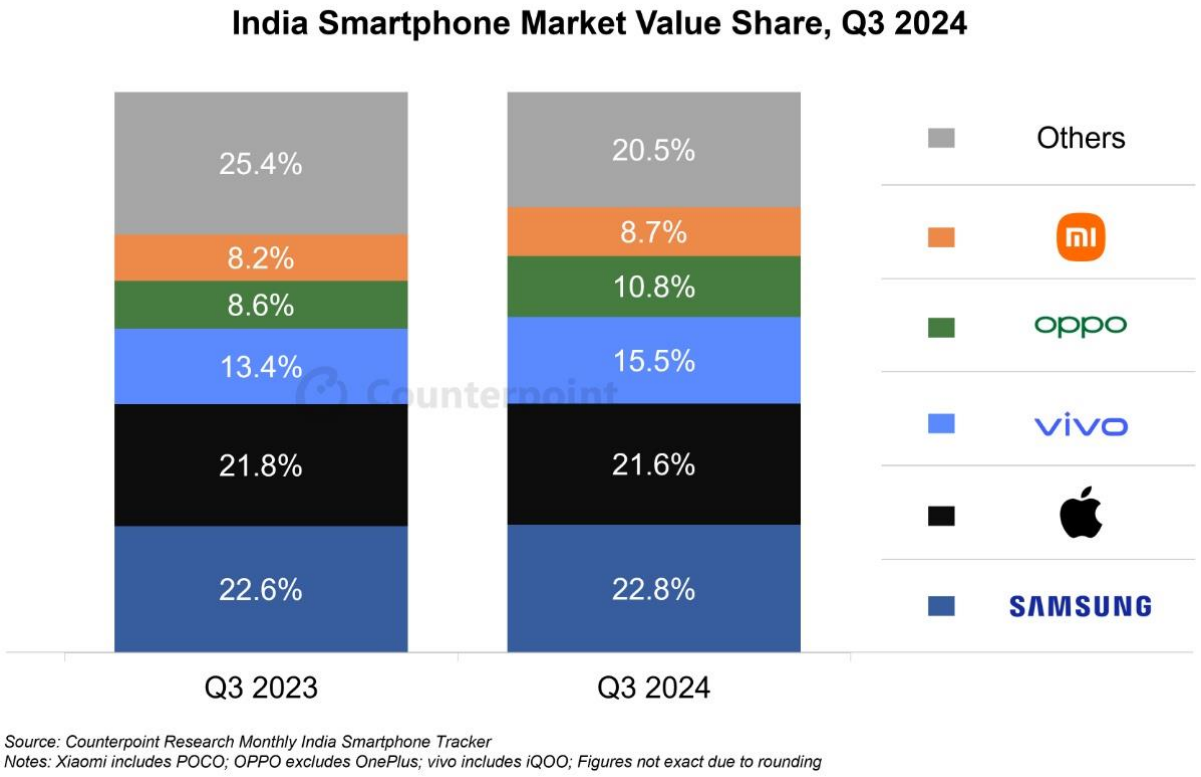


Apple India Sales Over the Years thought the visual of the graph identity.

Scenario 1: Market Penetration and Sales Trends

The visualization tool allows users to analyze iPhone sales data over time, across different regions in India. This includes examining trends in market penetration and identifying periods of high sales. Such insights help stakeholders understand how iPhone adoption has grown and where it has been most successful.

Apple has achieved a record-breaking presence in India, with its market share growing from 1% in 2019 to nearly 8% in 2025 and record high-value shares of 28% in the smartphone market. Driven by premiumization, local manufacturing, and festive,, data-driven strategies, the iPhone 16 became a top-selling device in 2025.



Market Penetration and Sales Trends in India

- **Rapid Growth:** Apple's market share in India grew from just 1% in 2019 to nearly 8% in 2025.
- **Record Sales & Value:** Apple hit record sales in late 2025, driven by strong demand for iPhones, Macs, and iPads. The company captured a record 28% value share of the Indian smartphone market.
- **Model Performance:** The iPhone 16 emerged as a highest-selling smartphone in 2025. Historically, models like the iPhone 8 Plus (64GB) were also highly rated and popular.
- **Key Drivers:** Growth is fueled by festive, discounts,, trade-in offers, and expanded local manufacturing.

- **Target Audience:** While historically seen as too expensive, increased affordability via EMI and retail expansion has expanded the user base, particularly among premium-conscious consumers.

Data-Driven Exploration via Tableau/Analytics

- **Visualization Goals:** Projects using Tableau (such as "iRevolution") utilize data sets from sources like Kaggle to create dashboards featuring KPIs like sales trends, user preferences, and regional performance.
- **Key Metrics:** Dashboards commonly track total units sold, revenue, average sales price, and customer rating trends, allowing for visual analysis of which models dominate the market.
- **Key Findings:** Analysis indicates a negative correlation between price and sales volume (lower-priced models sell better) but a positive correlation between discount percentages and customer ratings.

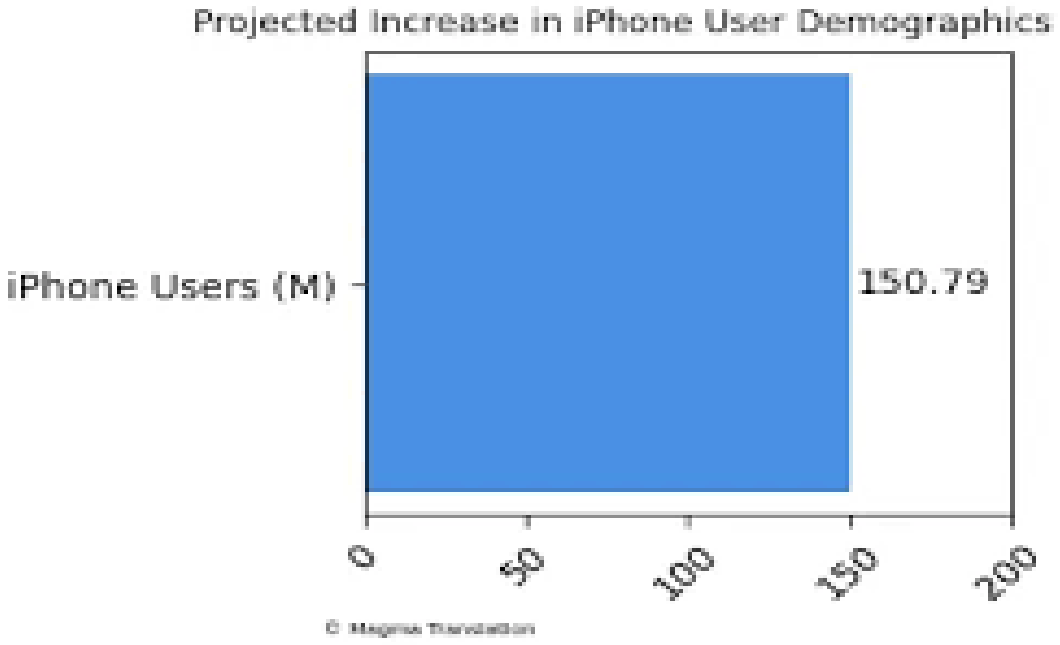
Strategic Insights

- **Premiumization Trend:** Consumers are increasingly moving towards high-end devices, allowing Apple to gain significant value share.
- **Retail Strategy:** The establishment of official Apple Stores and expanded retail, has been crucial to the recent surge in market penetration.

Scenario 2: User Demographics and Preferences

The project uses Tableau to explore the demographics of iPhone users in India, including age groups, income levels, and geographic distribution. This analysis can reveal which segments of the population are adopting iPhones and how user preferences vary across different regions, informing targeted marketing and product development strategies.

Apple's iPhone has achieved record-breaking success in India, with the iPhone 16 becoming the top-shipped model in 2025, capturing 9% volume and 28% value market share.Utilizing Tableau, data-driven analysis indicates a shift toward premiumization, with a young, mobile-first user base driving demand for high-end models, supported by increased local manufacturing.

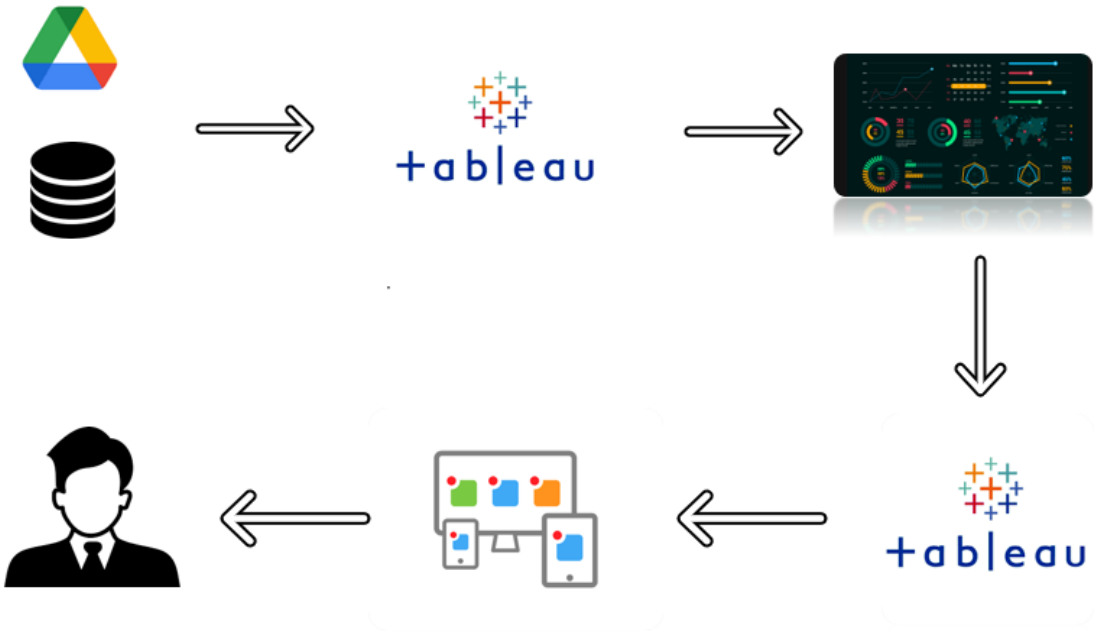


Key Findings on User Demographics and Preferences in India

- **Market Dominance:** Apple reached its highest-ever market share in India in 2025 by volume (9%) and value (28%), with 6.5 million iPhone 16 units sold in the first 11 months.
- **Demographic Profile:** The user base is significantly represented by Gen Z (born 1995–2012), who are mobile-first, using smartphones as their primary device for content and communication.
- **User Preferences:**
    - **Premium Demand:** A strong preference for the latest iPhone models (e.g., iPhone 16), showcasing a trend towards premium devices.
    - **Data-Driven Insights:** Analysis using Tableau shows that model-wise and region-wise performance is key to understanding consumer behavior and adapting marketing strategies.
- **Strategic Factors:** The "iRevolution" Tableau project highlights that Apple's success is bolstered by a "Make in India" initiative, aiming to produce a majority of iPhones locally by 2026.
- **AI Integration:** While specific consumer-facing AI data for India is evolving, the broader Indian market shows 74% AI adoption in IT, with 50% reporting benefits, suggesting a tech-savvy user base.

Scenario 3: Cultural and Social Media Impact

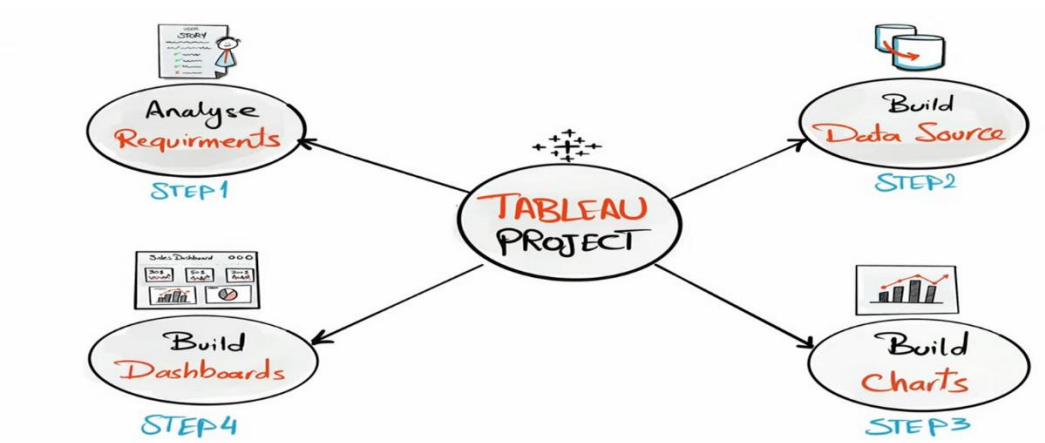
Through data visualization, the project assesses the cultural and social media impact of iPhone adoption in India. By analyzing sentiment and conversations on platforms like Twitter and Instagram, users can understand how the iPhone influences trends, lifestyles, and aspirations within Indian society. This insight helps stakeholders gauge brand perception and its role in shaping cultural narratives



ProjectFlow:

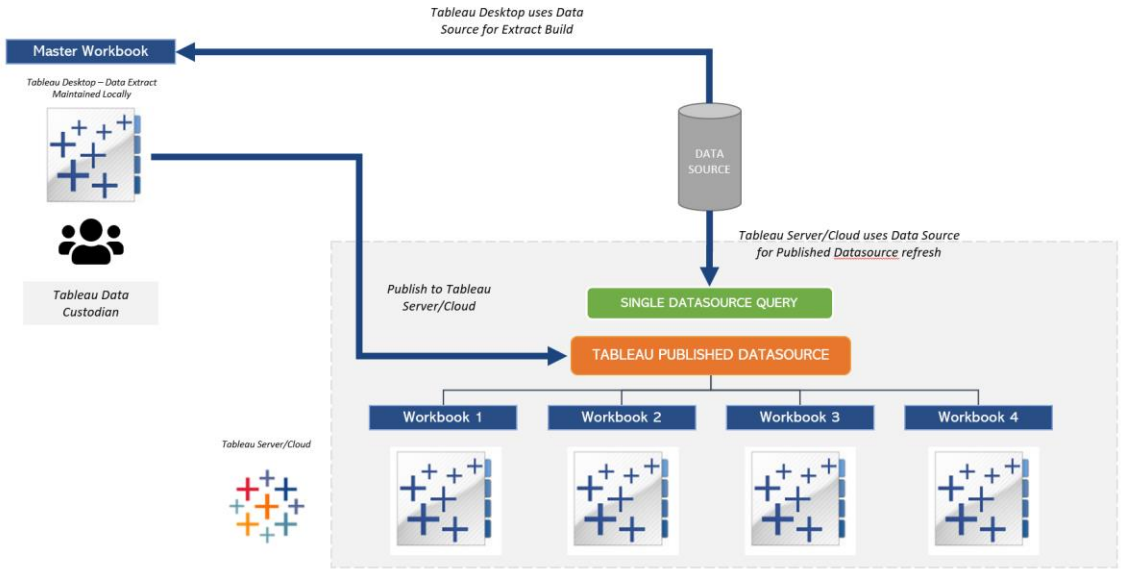
To accomplish this, we have to complete all the activities listed below,
?DefineProblem /Problem Understanding
o Specify the business problem
o Business requirements
o Literature Survey
o Social or Business Impact.
? Data Collection & Extraction from Database
o Collect the dataset,
o Storing Data in DB2
o Perform SQL Operations
o Connect DB2 with Cognos
? Data Preparation
o Prepare the Data for Visualization
? Data Visualizations
o No of Unique Visualizations
? Dashboard
o Responsive and Design of Dashboard
? Story
o No of Scenes of Story
? Report
o No of Visualization with detail information
? Performance Testing

o Amount of Data Rendered to DB2
o Utilization of Data Filters
o No of Calculation Fields
o No of Visualizations/ Graphs
? Web Integration
o Dashboard, Report and Story embed with UI With Flask
? Project Demonstration & Documentation
o Record explanation Video for project end to end solution
o Project Documentation-Step by step project development procedure



Data Collection & Extraction from Database

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes and generate insights from the data.



To analyze the iPhone's impact in India using Tableau, follow these streamlined steps for data collection, extraction, and visualization.

1. Data Collection Sources

Gather data from multiple dimensions to build a comprehensive view of the market:

- Public Datasets: Use curated datasets from Kaggle (e.g., "Apple products" or "mobile sales" datasets) containing specifications, pricing, and reviews.
- Market Research Reports: Consult Counterpoint Research or IDC for shipment volumes, market share (e.g., Apple's 28% value share in 2025), and top-selling models like the iPhone 16
- E-commerce Scraping: Extract real-time trends (ratings, discounts, and regional popularity) from platforms like Flipkart or Amazon India using Python tools like Pandas before importing into Tableau.
- Supply Chain Data: Monitor manufacturing shifts, such as India's increasing contribution to global production (expected to reach 20% in 2025) via official trade and manufacturing portals.

2. Data Extraction and Connectivity

Connect Tableau to your chosen data sources to prepare for analysis:

- Database Connectivity: Tableau supports native connectors for SQL Server, MySQL, and cloud databases like Google BigQuery.

- Live vs. Extract Connections:
    - Live Connection: Best for real-time tracking of sales campaigns or inventory.
    - Data Extracts (.hyper): Recommended for large datasets (millions of rows) to improve performance and enable offline analysis.
- Tableau Prep: Use Tableau Prep Builder to clean, shape, and join data from different sources (e.g., merging sales data with manufacturing stats) before visualization.

3. Key Metrics to Visualize

Focus on these performance indicators to demonstrate market impact:

- Market Share Trends: Visualize the leap from 8% to 25% share in global shipments since 2022.
- Price-Rating Correlation: Analyze how affordability and discounts on platforms like Flipkart influence customer engagement and ratings.
- Regional Performance: Use Tableau's geographic mapping to track iPhone adoption in urban centers versus tier-2 cities.
- Year-over-Year (YoY) Growth: Compare sales volumes between generations, such as the iPhone 17

's 15-20% higher initial sales compared to previous models.

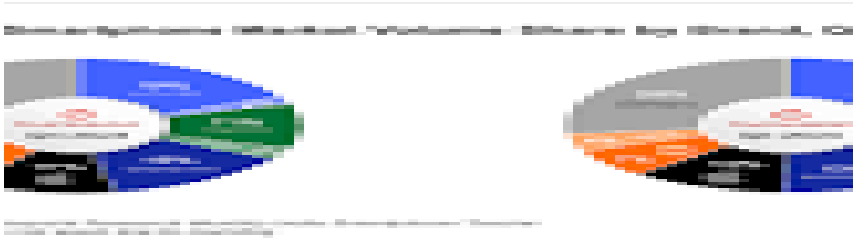
Would you like a step-by-step guide on creating calculated fields in Tableau to track iPhone market share variance?

- Final Report | PDF | Analytics | I Phone - Scribd

Final Report. The iRevolution project analyzes the impact of Apple's iPhone in the Indian market through interactive visual analyt...

- Over One in Every Five Smartphones Shipped in India is Now Premium

2 Feb 2026 — Apple led the market in 2025 in value terms with a 28% share, as festival-led promotions and deeper channel expansion supported re...

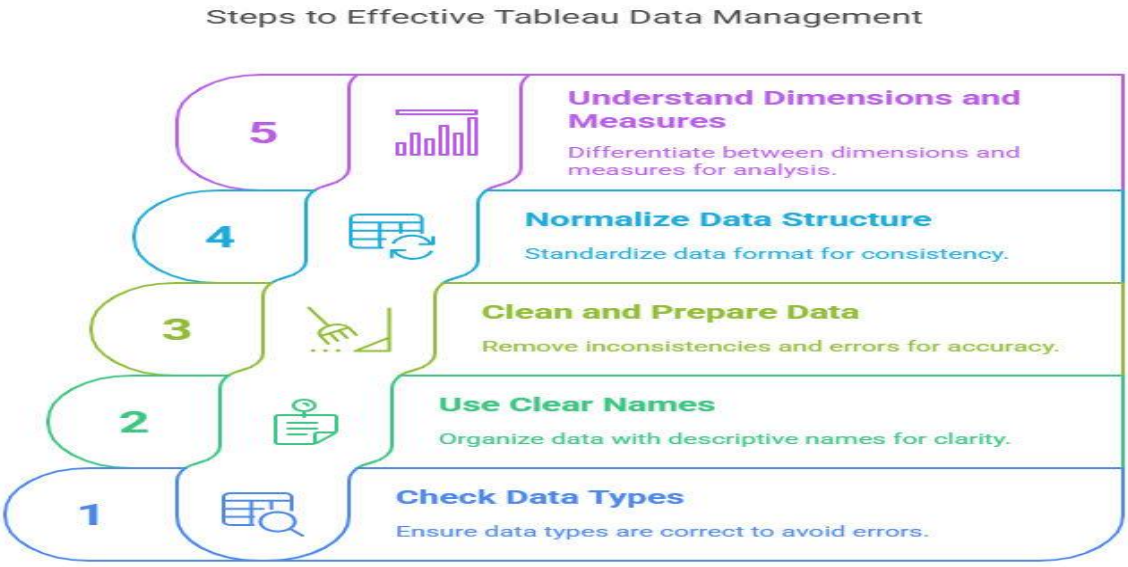


- The Made-in-India iPhone race that's reshaping Apple's global supply chain

13 Jun 2025 — Though China still produces over 75% of global iPhones, India now contributes about 18% and is expected to rise to 20% in 2025. De...

Data Preparation

This Mile stone explains about Data Preparation. Clean, transform, and organize the connected data to ensure consistency and accuracy. Create calculated fields, handle null values, and structure the data appropriately for effective visualization and insightful analysis in Tableau.



Prepare the Data for Visualization

In this step, we focus on preparing the dataset for visualization in Tableau. Fortunately, the dataset we're working with has already been pre-cleaned, meaning major cleaning steps such as handling missing values, removing duplicates, and correcting inconsistencies have already been taken care of.

However, even with a clean dataset, it's still essential to go through a brief review process to ensure it's truly ready for analysis:

- DataReview& Exploration
  While the dataset is clean, it's good practice to explore it briefly—checking data types, value ranges, and distributions. This helps us understand the structure, identify any potential outliers, and gain familiarity with the data we'll be visualizing.
- Filtering and Structuring for Purpose
  Depending on the business question, we may still need to filter the data to focus on specific subsets—such as certain time periods, regions, or product categories. Structuring the data to match the visualization goal helps ensure relevance and clarity.
- Field Renaming & Final Formatting
  To enhance clarity in Tableau, we ensure field names are intuitive and consistent. We also check for proper data types (e.g., date fields, numeric values) and relationships if the dataset spans multiple tables.
- Optional Calculated Fields
  If needed, we can create calculated fields (e.g., profit margin, growth rate) to support

- deeper analysis. Even with a clean dataset, these additions can make our visualizations more insightful.
- Validation for Accuracy
  Lastly, a quick validation against the source or summary metrics ensures everything is accurate. This final step helps maintain trust in the insights generated.

Collect the dataset

Please use the link to download the dataset:

https://docs.google.com/spreadsheets/d/1p1ZWaYcEuFl5UNFcmNvpkXi3JnoHamut/edit?usp=sharing&ouid=113247709954189786236&rtpof=true&sd=true

Data contains all the meta information regarding the columns described in the CSV files. We have provided the XLSX file:

Column Description for BigML_Dataset.csv:

The file apple_products.xlsx contains a total of 7 sheets. Each sheet corresponds to a different parameter related to iPhones/Smartphones.

The sheets are:

- apple_products.csv
- Flipkart_smartphone
- Annual revenue
- Market penetration (iPhone)
- Country wise share
- Quarterly-share
- Model-wise share

Based on common structures for market impact analysis, a dataset regarding iPhone impact in India (suitable for BigML and Tableau) typically includes columns analyzing sales trends, consumer demographics, and price points. Key columns often include **Region/City**, **Sales Volume**, **Price Point**, **Date**, **Model Name**, and **Demographic Info**.

Typical Column Descriptions:

- **Date (Date/Time):** The specific date of transaction or data collection (e.g., monthly).
- **Region (String):** Geographical location within India (e.g., Tier 1, Tier 2 cities, North/South/East/West).
- **Model Name (String):** The specific iPhone model (e.g., iPhone 13, iPhone 14 Pro).
- **Units Sold (Integer):** Total number of units sold per region/period.
- **Sales Price (Float/Currency):** The selling price of the iPhone in Indian Rupees (INR).
- **Consumer Age (Integer/Bucket):** Demographic of the buyer.
- **Marketing Spend (Float):** Localized marketing investment for the period.
- **Competitor Price (Float):** Price of rival flagship phones (e.g., Samsung Galaxy S-series) to measure market impact.

This data allows for identifying trends in regional demand, price sensitivity, and the growth rate of Apple's market share in India, which can be visualized in Tableau for insights.

Data Visualization

Data visualization is the process of creating graphical representations of data in order to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

The "iRevolution" project on Tableau Public analyzes Apple's iPhone impact in India, using interactive dashboards to visualize sales trends, market performance, and user preferences. It utilizes dark-themed, data-driven storytelling to help stakeholders make informed decisions on market penetration. The analysis covers key performance indicators (KPIs) and growth opportunities.

Key Aspects of iPhone Impact Visualization in Tableau

- **Performance Metrics:** The dashboards track sales trends to highlight how Apple has strengthened its standing in the competitive Indian smartphone market.
- **User Preferences & Market Share:** Visuals show the growth in user base and the popularity of specific iPhone models.
- **Interactive Dashboards:** The project uses interactive elements to allow stakeholders to explore data, such as sales trends and user behavior, directly.
- **Data-Driven Decision Making:** The visualization helps in identifying key growth opportunities, monitoring engagement, and optimizing strategies for the Indian market.

Components of the Tableau Analysis

- **Data Sources:** The analysis likely involves data on smartphone sales, consumer demographics, and brand positioning in India.
- **Visualization Types:** The project uses various charts, such as trend lines, bar charts, and maps, to represent the data visually.
- **Impact Assessment:** The dashboards aim to show the "iRevolution" by analyzing how Apple's market share has changed over time.

For more, the full project can be viewed on Tableau Public.

No of Unique Visualizations

The number of unique visualizations that can be created with a given dataset. Some common types of visualizations that can be used to analyze the performance and efficiency of a project include bar charts, line charts, heat maps, scatter plots, pie charts, Maps, etc. These visualizations can be used to compare performance, track changes over time, and show distribution, and relationships between variables.

Based on analysis of "iRevolution" and similar Tableau public projects exploring Apple's iPhone impact in India, visualizations generally focus on analyzing market share, sales, and consumer trends through several unique lenses.

Key unique visualizations and analytical dimensions include:

- **Regional Sales Performance Maps:** Visualizing the sales distribution of iPhone models across different Indian states.
- **Price Point vs. User Rating Scatter Plot:** Analyzing the negative correlation between iPhone price and popularity, or positive correlation between discounts and sales volume.

- **Quarterly Growth Trend Lines:** Tracking Apple's quarterly revenue and market share growth (e.g., reaching 9% volume share by 2025).

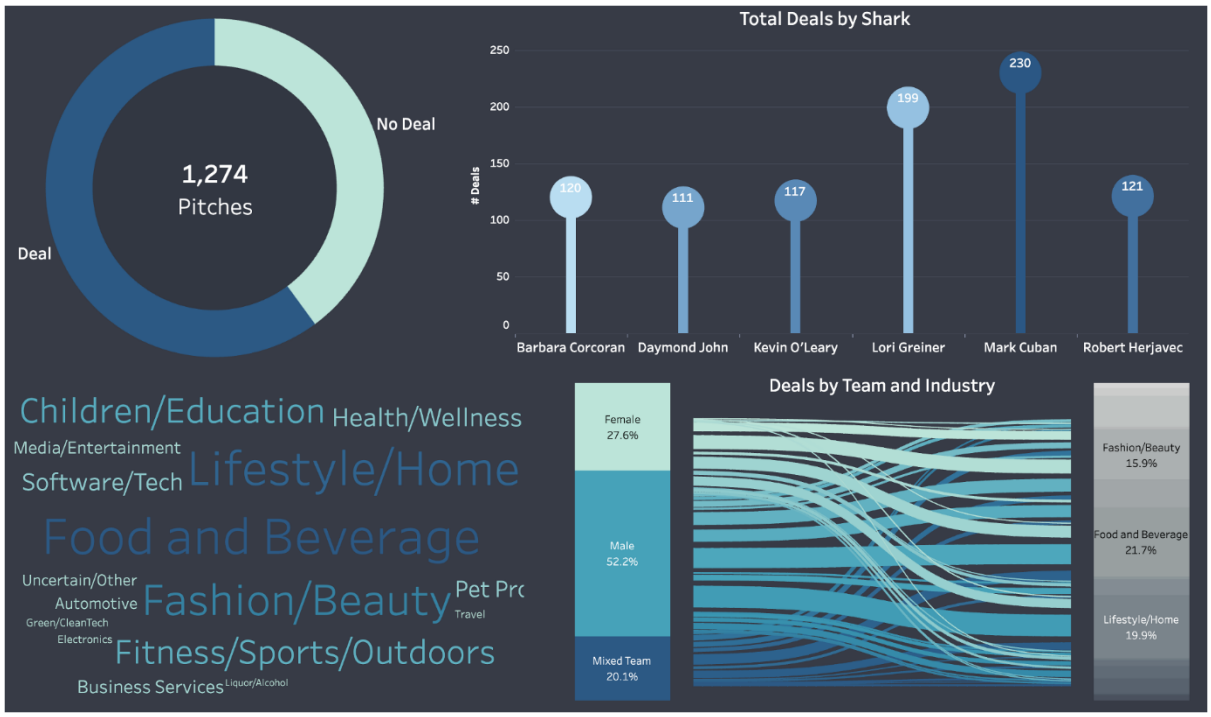
- **Model-Specific Adoption Bar Charts:** Comparing the popularity of different models (e.g., iPhone 16/17 series).

- **EMI/Financing Trend Charts:** Visualizing the high percentage (80%) of iPhones bought on EMI to show financial accessibility.

- **Manufacturing Output/Export Growth:** Tracking the rise in "Make in India" iPhone exports ($50B+ milestone).

- **Average Selling Price (ASP) Analysis:** Monitoring shifts in consumer preferences toward premium models.

The "iRevolution" project specifically emphasizes a **dark UI dashboard** to present these metrics interactively.



KPI:

KPI

Brand	Discount Percent..	Mrp	Star Rating	Sale Price	Sales difference
Apple	0	77,000	5	77,000	0

Analyzing the impact of the iPhone in India using **Tableau** allows you to transform complex sales and manufacturing data into actionable insights. Projects like iRevolution leverage these visual analytics to track the shift from a niche luxury product to a mainstream market leader.

Key Performance Indicators (KPIs) to Track

To measure the impact effectively, focus on these primary metrics:

- **Market Share Growth:** Monitor Apple's climb from 1% in 2019 to nearly **8% by late 2025**.
- **Sales Volume by Model:** Track high-performers like the **iPhone 16** which sold approximately 6.5 million units in its first 11 months in India.
- **Manufacturing Output:** Measure the value of iPhones produced in India, which jumped 60% to **$22 billion** in the year ending March 2025.
- **Regional Adoption:** Visualize regional sales trends (Urban vs. Rural) and identify top-performing states.
- **Average Selling Price (ASP) & Discounts:** Analyze how festive discounts and EMI offers drive volume despite premium pricing.

Building the Dashboard in Tableau

1. **Data Integration:** Connect sources like Statista, market research reports (Counterpoint, IDC), or manufacturing datasets.
2. **Calculated Fields:** Use Tableau Calculated Fields to create "Year-over-Year (YoY) Growth" or "Current Year vs. Prior Year" benchmarks.

3. **Visual Best Practices:**

KPI Tiles: Place high-level numbers (Total Sales, Market Share %) at the top of the dashboard for immediate impact.

Geographic Maps: Use maps to highlight sales density across Indian states.

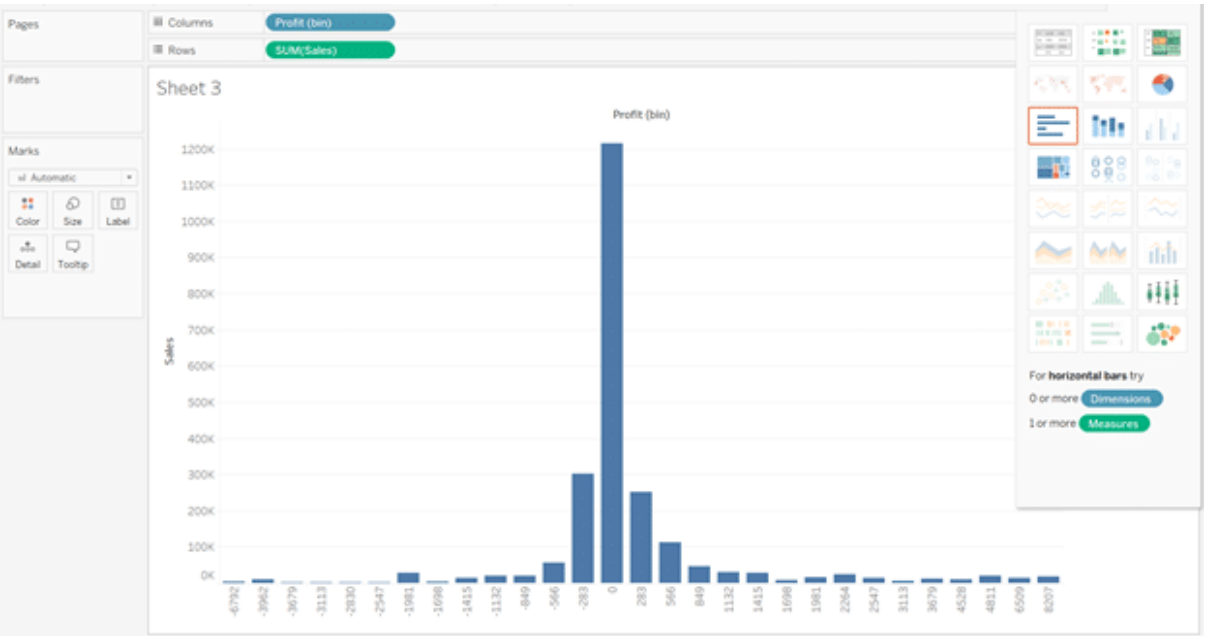
Trend Lines: Use dual-axis line charts to compare production value against domestic sales volume.

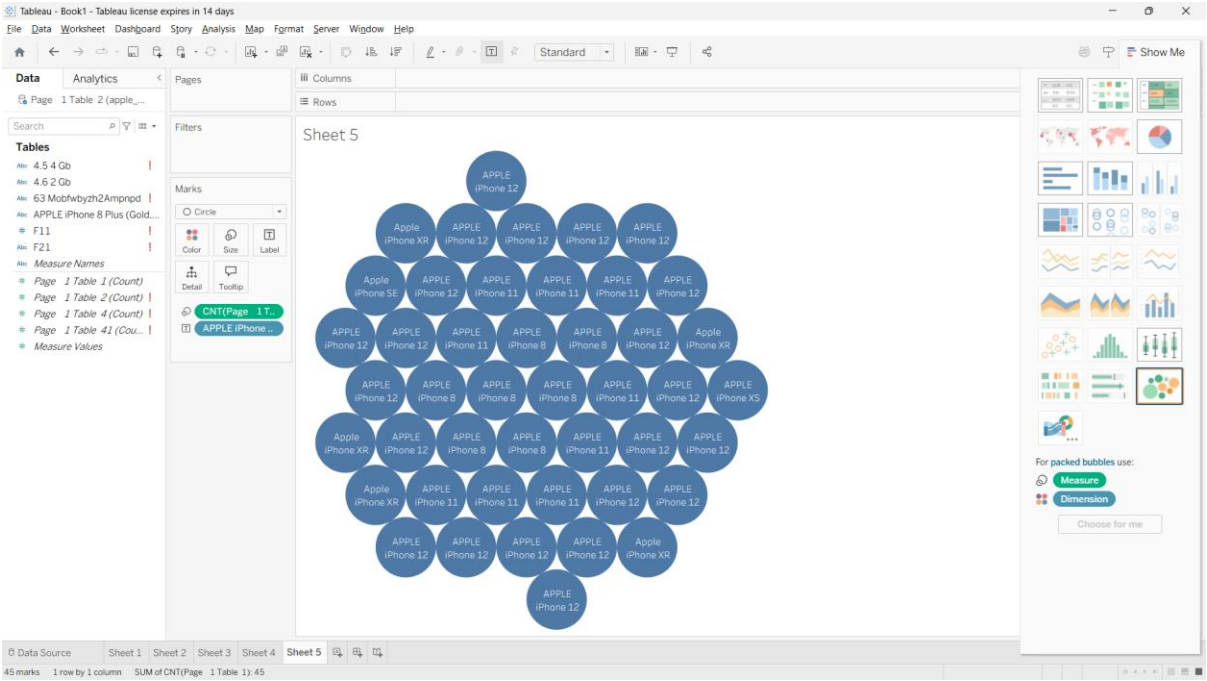
Color Coding: Use green/red indicators to show if current performance meets or falls below set targets.

Strategic Insights for 2026

- **Production Shift:** Apple aims to shift assembly of **all US-sold iPhones** (over 60 million annually) to Indian factories by the end of 2026.
- **Adoption Rate:** The **iPhone 17** set a record in late 2025, accounting for 57% of all iPhones sold in its launch phase.

Would you like a step-by-step guide on creating a specific **Calculated Field** for YoY growth or a list of **data sources** to populate your dashboard?.



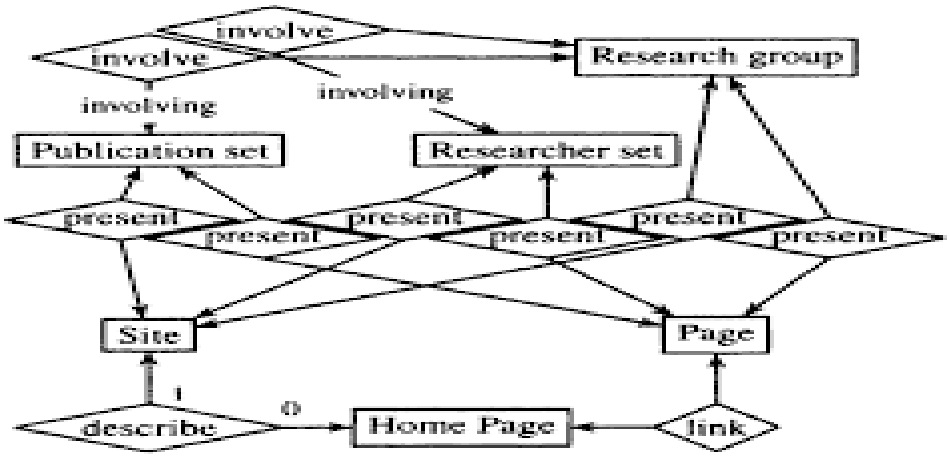


S

Model Specification

Analyzing the impact of iPhone model specifications in India using Tableau involves visualizing how technical features (like storage, battery, and 5G) correlate with market performance and consumer sentiment.

A rigorous, versioned document that defines how AI models should behave. Acts as a source of truth for model behavior, evaluations, documentation, and even code generation. It outlines principles, rules, and trade-offs that guide model responses, tone, personality, and ethical boundaries.



Sheet 2

Model	Processor	Front Ca..	Rear Camera	Col..	
APPLE IPHONE 11	A Bionic Chip	12MP	12MP + 12MP	Black	90,900
APPLE IPHONE 12	A Bionic Chip with Next Generation Neural Engine	12MP	12MP + 12MP	Black	199,700
				Blue	59,900
APPLE IPHONE 12 MINI	A Bionic Chip with Next Generation Neural Engine	12MP	12MP + 12MP	Blue	74,900
				Black	74,900
APPLE IPHONE 13	A Bionic Chip	12MP	12MP + 12MP	Blue	149,800
APPLE IPHONE 14	A Bionic Chip, Core	12MP	12MP + 12MP	Blue	189,800
APPLE IPHONE 14 PLUS	A Bionic Chip, Core	12MP	12MP + 12MP	Blue	99,900

Analyzing the impact of iPhone model specifications in India using Tableau involves visualizing how technical features (like storage, battery, and 5G) correlate with market performance and consumer sentiment.

Key Visualizations for iPhone Analysis

- **Market Share & Value:** As of early 2026, Apple holds a record **28% value share** in India's smartphone market [27]. Use a **Filled Map** to visualize this regional performance across Indian states [6].
- **Specification Performance:** Compare features like the **A16/A17/A19 Bionic chips** against sales data. High-end models like the

iPhone 15 Pro Max

(4422 mAh battery, 120Hz display) and

iPhone 17 Pro

(3000 nits brightness) drive the "premiumisation" trend in India [3, 10, 27].

- **Price vs. Features:** A **Scatter Plot** can help visualize the "Price-Performance Trend Line." For example, the

iPhone 13

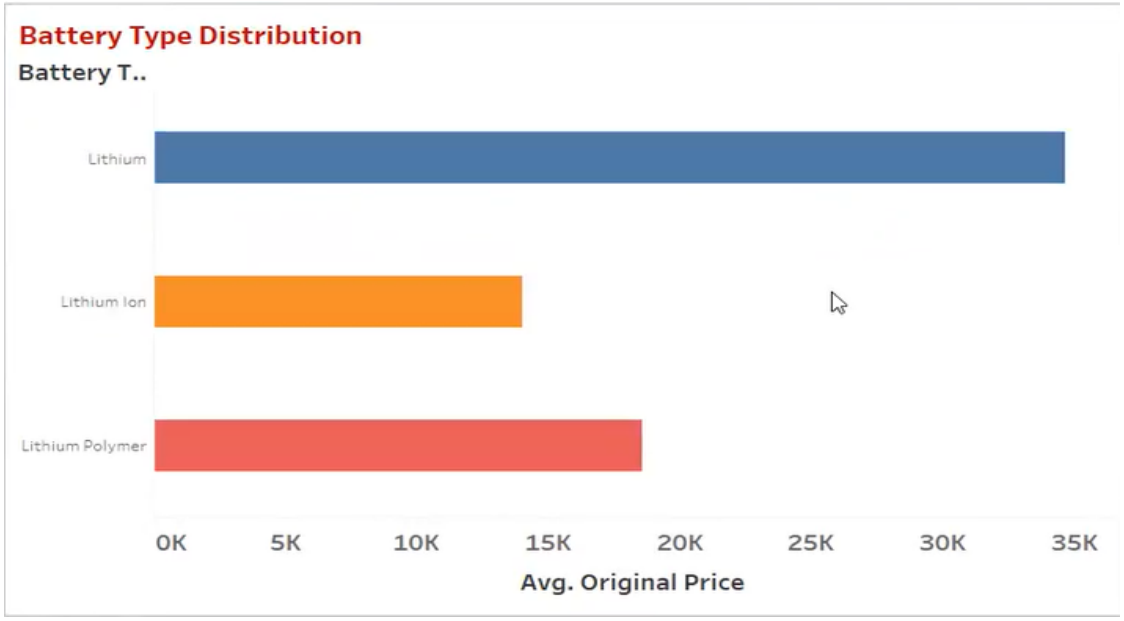
(starting ~₹41,000) offers high value compared to the

iPhone 17 Pro

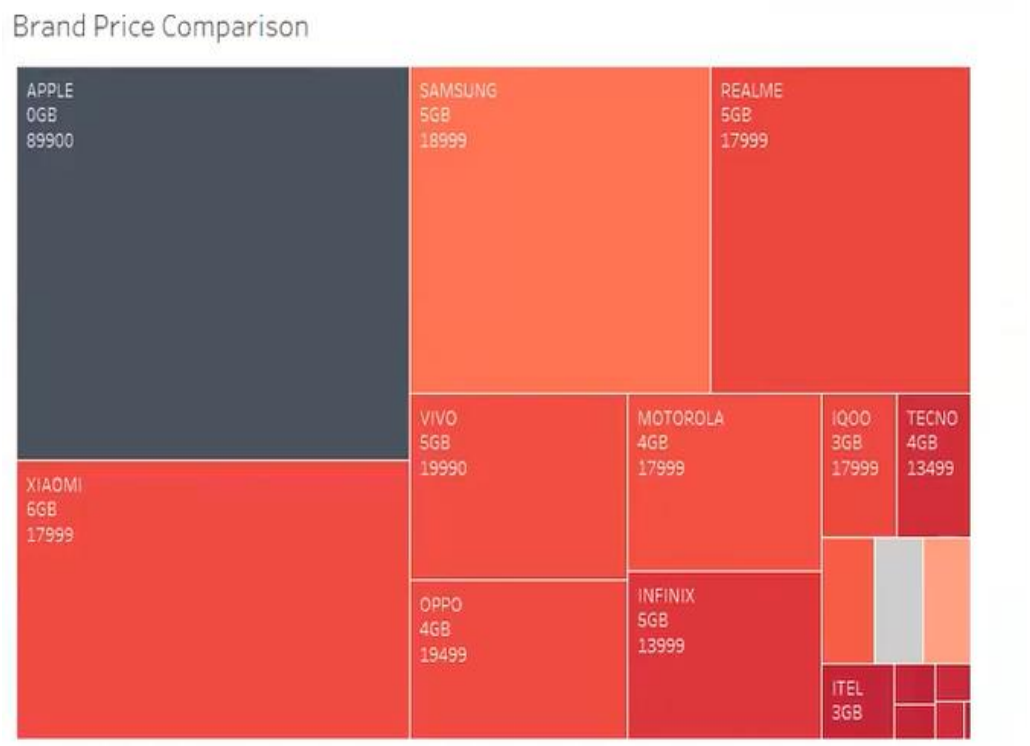
(~₹1,32,490) [2, 10, 19].

- **Sentiment Analysis:** Use **Word Clouds** or **Bar Charts** to visualize user reviews. Positive Indian consumer sentiment is often tied to **long battery life** and **display quality**.

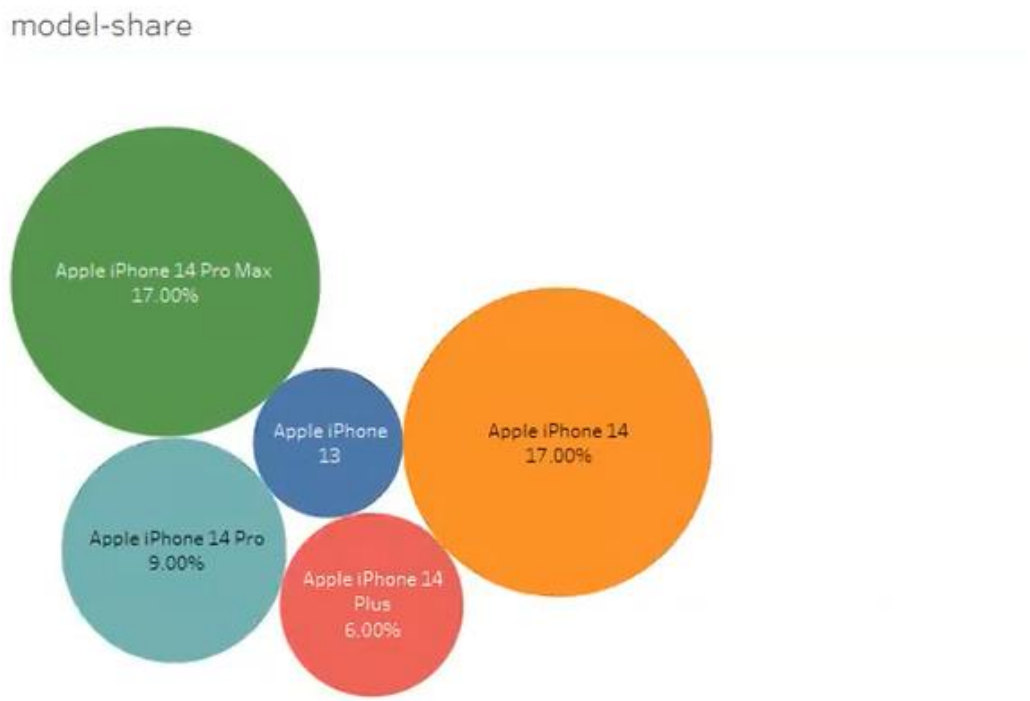
BarChart showing Battery-Type



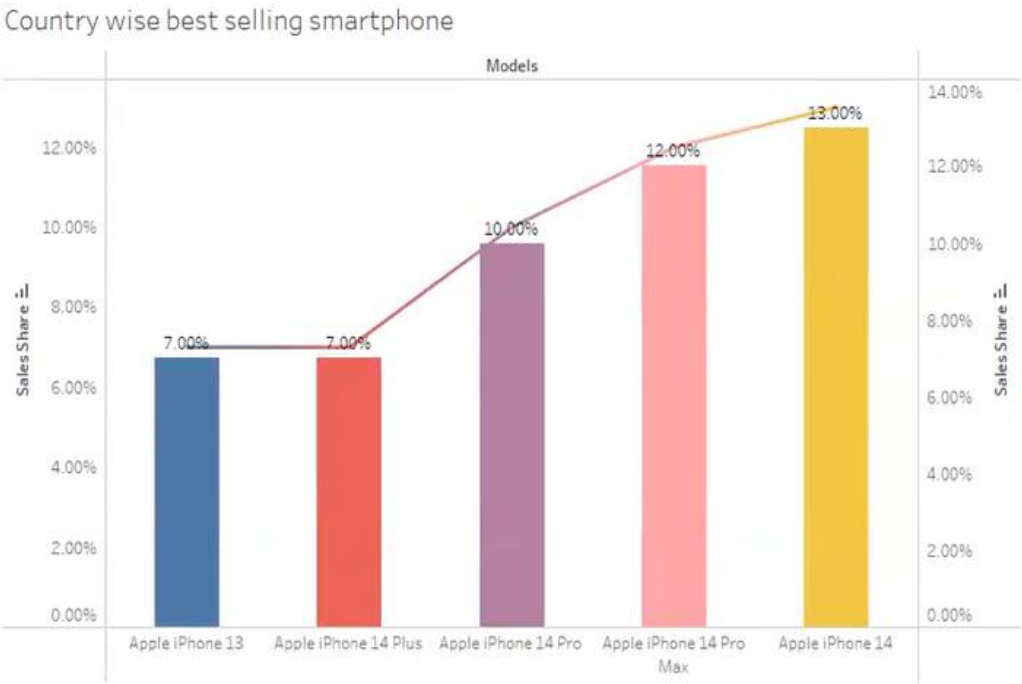
Treemap showing Brand- Price Comparison



Bubble Chart showing Model- Wise Share of iPhone



Lined Bar-Chart showing Country-Wise Best Selling Smartphone



Map Showing Global Market Share

Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data, and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

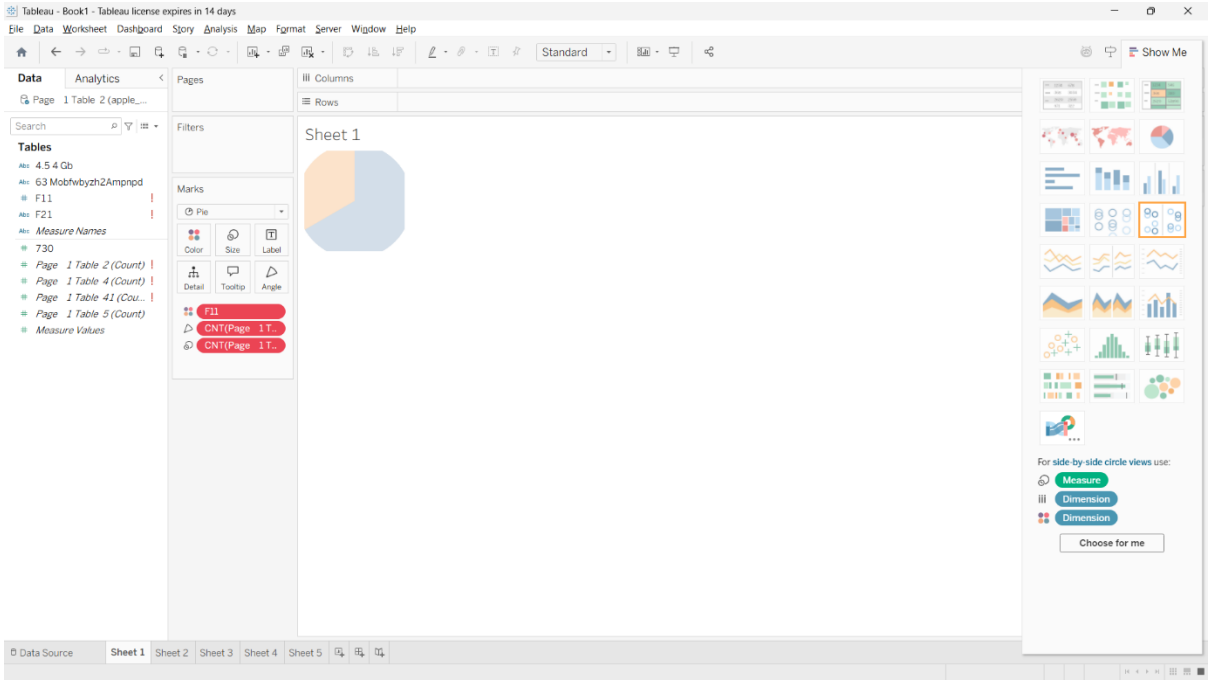
Story

A data story is a way of presenting data and analysis in a narrative format, with the goal of making the information more engaging and easier to understand. A data story typically includes a clear introduction that sets the stage and explains the context for the data, a body that presents the data and analysis in a logical and systematic way, and a conclusion that summarizes the key ?ndings and highlights their implications. Data stories can be told using a variety of mediums, such as reports, presentations, interactive visualizations, and videos.

No of Scenes of Story

The number of scenes in a storyboard for iRevolution will depend on the complexity of the analysis and the specific insights that are trying to be conveyed. A storyboard is a visual

representation of the data analysis process and it breaks down the analysis into a series of steps or scenes.



Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

Amount of Data Rendered to Tableau

Monitor the volume of data being pulled and rendered from the database to ensure queries are optimized and not overloading the system.

The amount of data that is rendered to a database depends on the size of the dataset and the capacity of the database to store and retrieve data.

Utilization of Filters

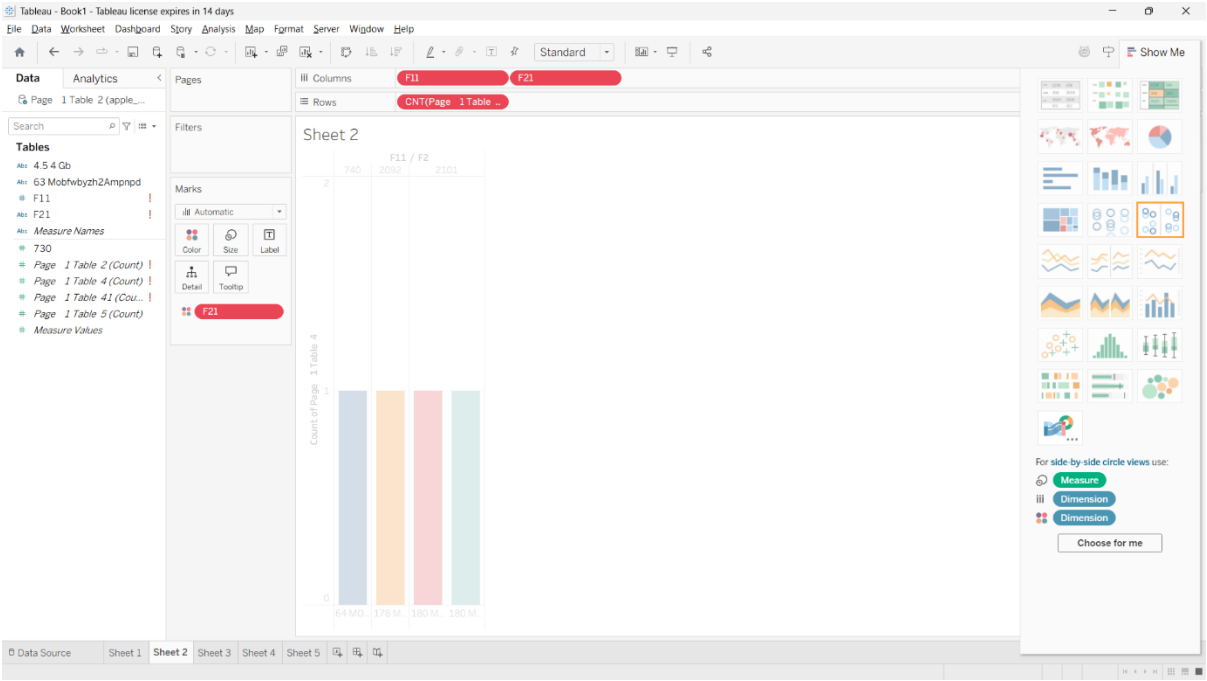
Filters are essential devices and methods used to remove unwanted particles, contaminants, or specific frequencies from liquids, gases, electronic signals, or data sets. They ensure purity, safety, and efficiency across industries like water treatment, automotive, and pharmaceuticals, while enhancing signal quality in communication and electronics.

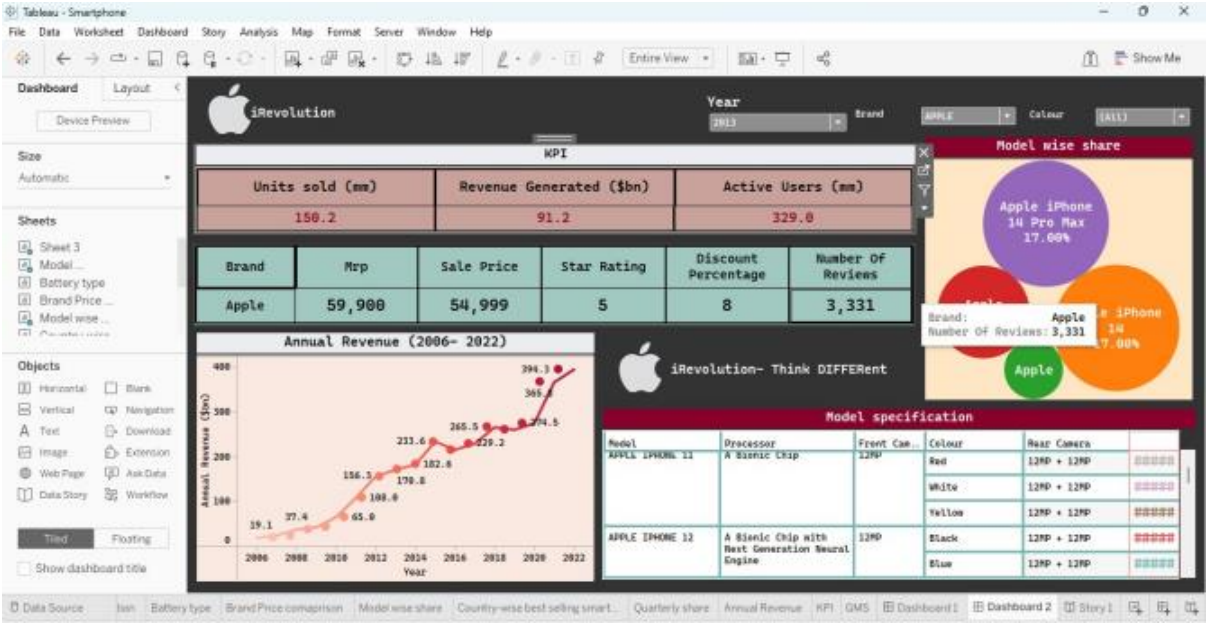
Key Utilizations of Filters:

- **Fluid & Air Purification:** Filters remove contaminants from water (sand, bacteria) and air (dust, smoke) to improve quality and safety.
- **Industrial Applications:** Used in the oil and gas industry to remove impurities, protecting equipment from damage. In food processing, they remove unwanted substances from liquids like milk or juice.
- **Electronics & Signal Processing:** Electronic filters (active/passive) remove noise from signals, smooth data, and manage frequencies in communication systems, AI, and power grids.
- **Healthcare:** Filters are used in breathing apparatuses and to ensure sterile environments, including the use of 0.2-µm in-line filters to reduce hospital-acquired bacteremias.
- **Data Analysis & Software:** The {Link: FILTER function in Excel https://www.youtube.com/watch?v=ywwUqDQPqQg} dynamically extracts data based on specific criteria. Similarly, JavaScript's {Link: Array.prototype.filter() https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/filter} creates new arrays from existing data based on set conditions.
- **Photography:** Lens filters are placed on cameras to enhance, color-correct, or protect lenses.

Benefits of Proper Filter Utilization:

- **Improved Efficiency:** Reduces equipment failure and maintenance costs.
- **Quality Control:** Ensures the integrity of products in pharmaceuticals and manufacturing.
- **Environmental Protection:** Cleans exhaust and waste streams.



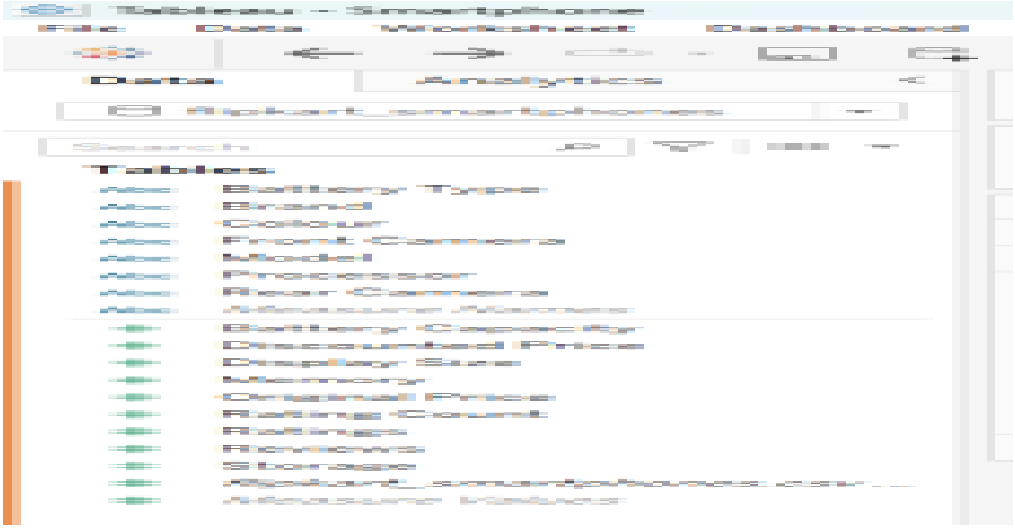


No of Calculation Fields

A calculated field is like making your own custom column in Tableau. Instead of only using the dataset values as they are, you can create new values using formulas.

Step-by-Step:

1. Open Tableau and connect your dataset.
2. In the Data pane (left side where all fields are listed), Right-click ? choose Create Calculated Field.
3. A dialog box opens ? give it a Name
4. Write your formula in the editor



No of Visualizations/ Graphs

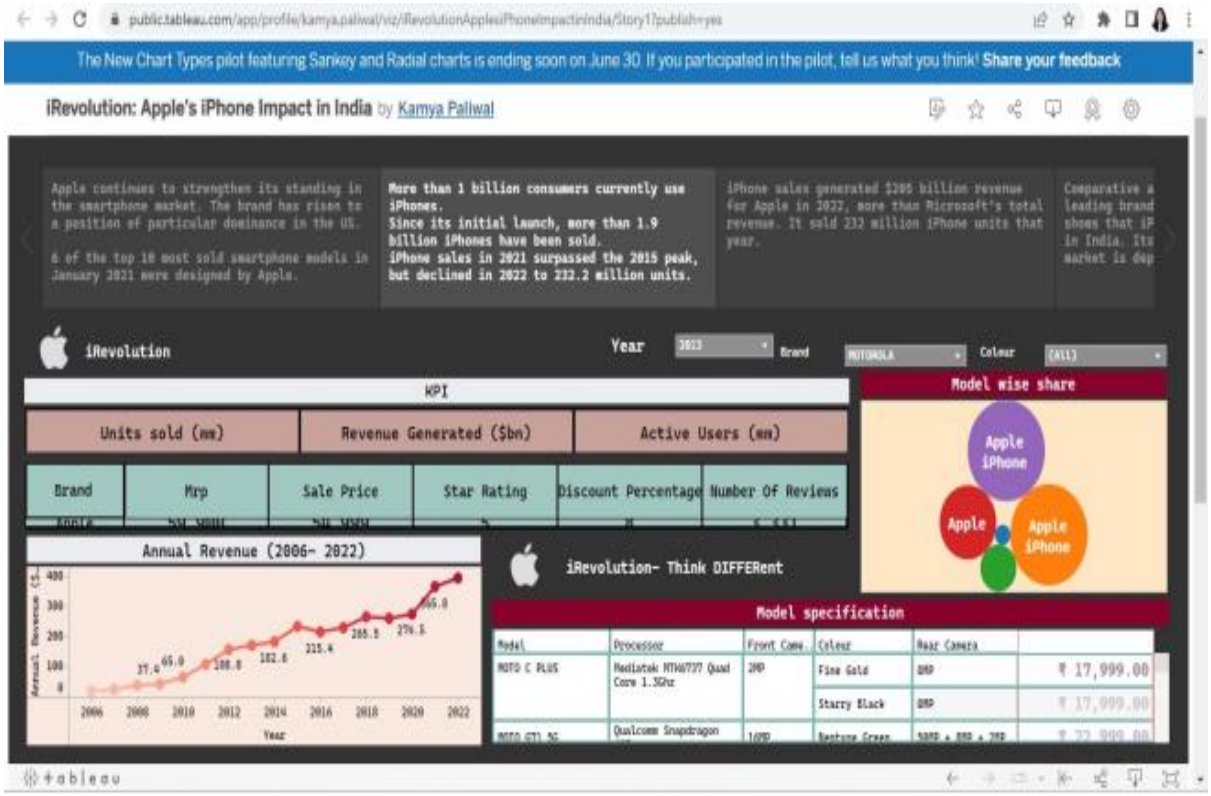
- KPI
- Model Speci?cation
- Model- Wise share
- Battery-Type distribution
- Brand- Price Comparison
- Model- Wise Share of iPhone
- Country-Wise Best-Selling Smartphone
- Annual Revenue Year-Wise
- Global Market Share

Web integration

- Web integration of a Tableau Dashboard Story involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It

supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

Story:



S

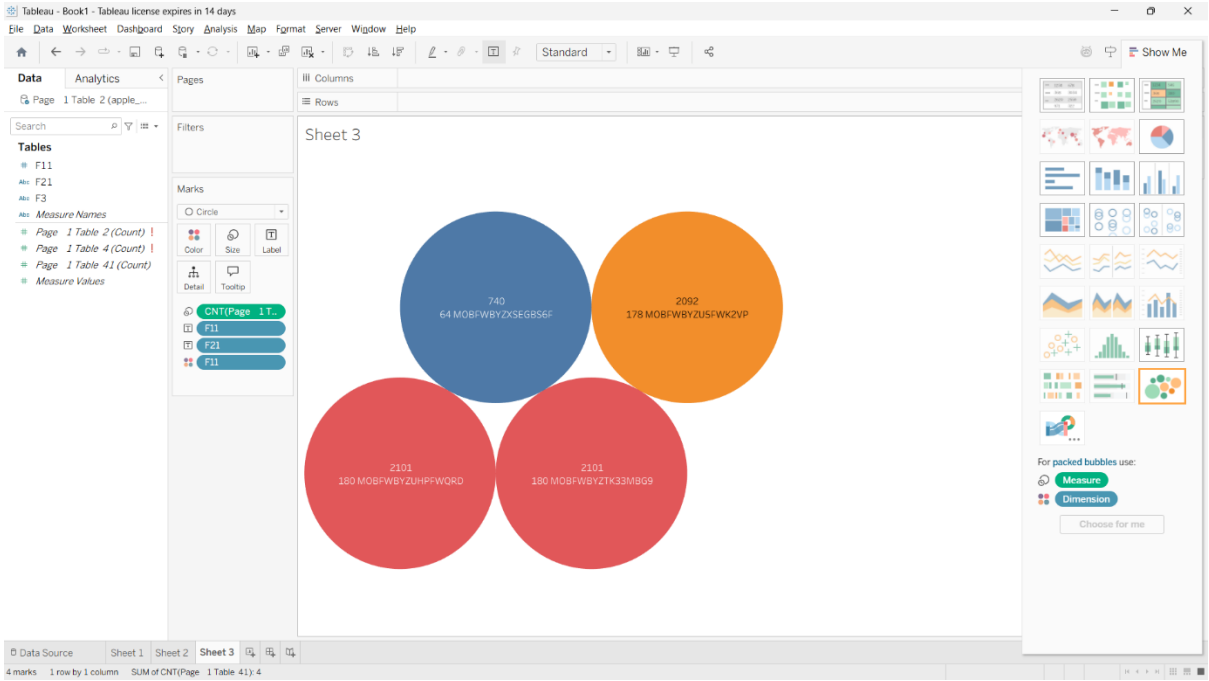


Tableau - Book1 - Tableau license expires in 14 days
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Data
Analytics
Tables
F11
F21
F3
Measure Names
Page 1 Table 2 (Count)
Page 1 Table 4 (Count)
Page 1 Table 41 (Count)
Measure Values
Pages
Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
CNT(Page 1 T...
F11
F21
F11
Columns
Rows
Sheet 3
740
64 MOBFWBYZXSEGBS6F
2092
178 MOBFWBYZUSFWK2VP
2101
180 MOBFWBYZUHPFWQRD
2101
180 MOBFWBYZTK33MBG9
Show Me
For packed bubbles use:
Measure
Dimension
Choose for me
Data Source
Sheet 1
Sheet 2
Sheet 3
4 marks
1 row by 1 column
SUM of CNT(Page 1 Table 41): 4

public.tableau.com/app/profile/kamya.paliwal/viz/iRevolutionApplesiPhoneImpactInIndia/Story1?publish=yes
The New Chart Types pilot featuring Sankey and Radial charts is ending soon on June 30. If you participated in the pilot, tell us what you think! Share your feedback
iRevolution: Apple's iPhone Impact in India by Kamya Paliwal
Apple continues to strengthen its standing in the smartphone market. The brand has risen to a position of particular dominance in the US.
6 of the top 10 most sold smartphone models in January 2021 were designed by Apple.
More than 1 billion consumers currently use iPhones.
Since its initial launch, more than 1.9 billion iPhones have been sold.
iPhone sales in 2021 surpassed the 2015 peak, but declined in 2022 to 232.2 million units.
iPhone sales generated $205 billion revenue for Apple in 2022, more than Microsoft's total revenue. It sold 232 million iPhone units that year.
Year
2006
2022
Annual Revenue ($bn)
400
300
200
100
0
19.1
24.4
37.4
42.9
65.0
108.0
156.3
170.8
182.6
233.6
215.4
229.2
265.5
260.1
274.5
265.0
304.3
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year
tableau

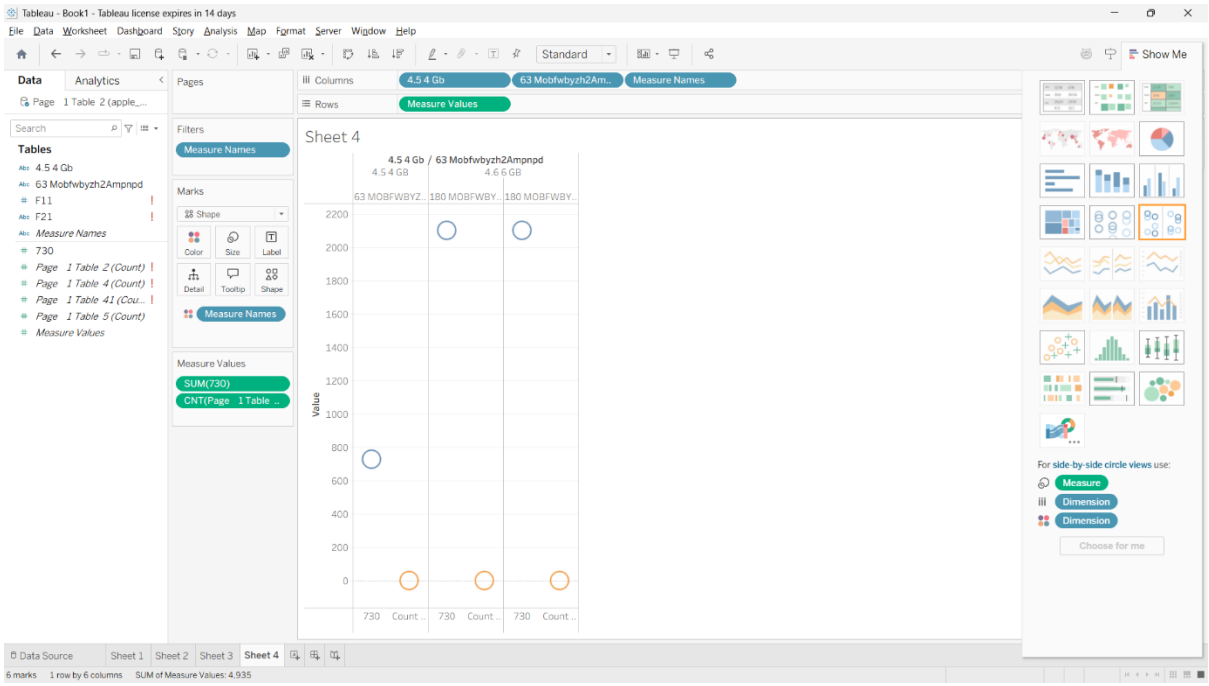
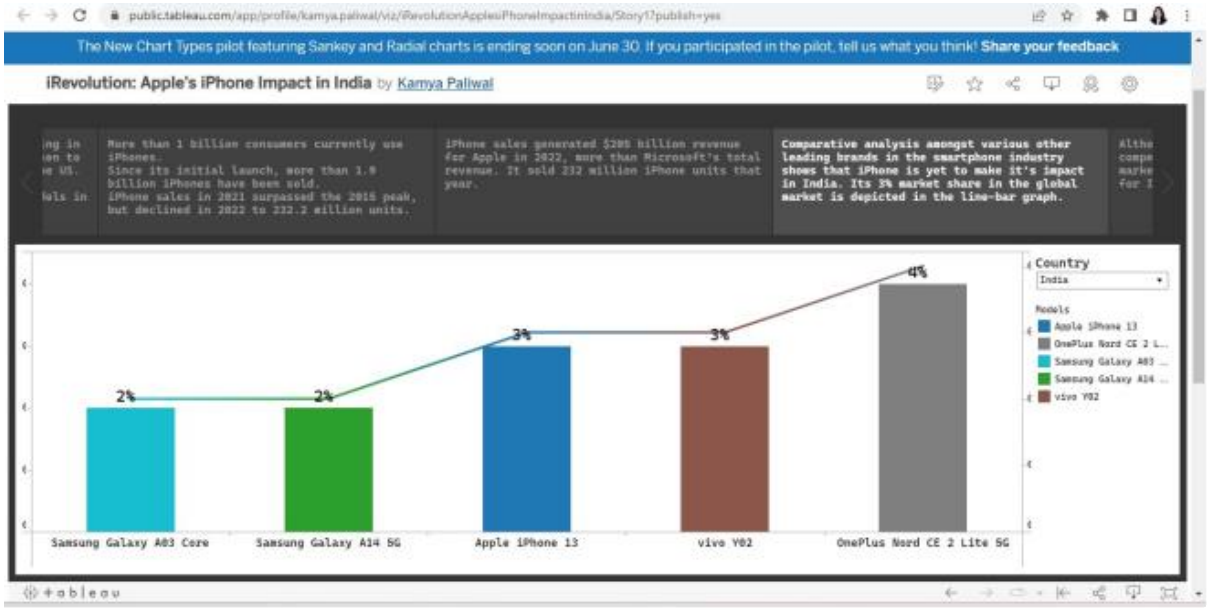
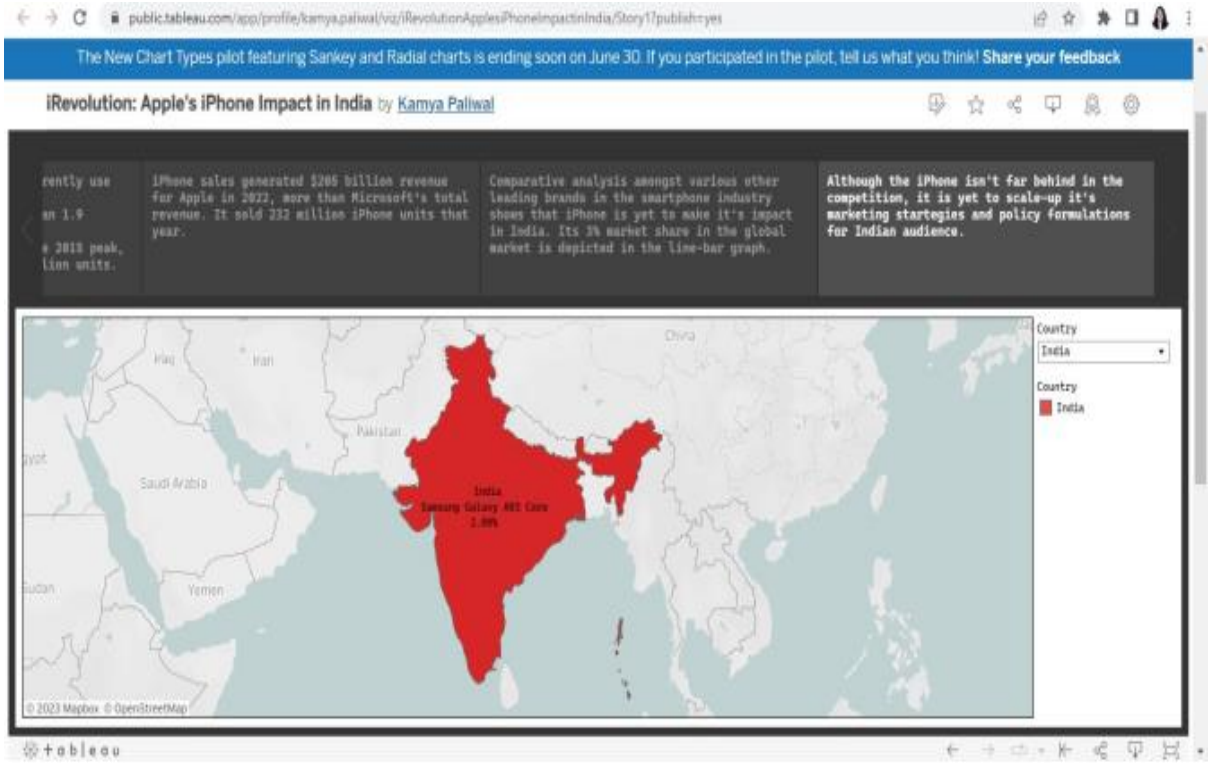
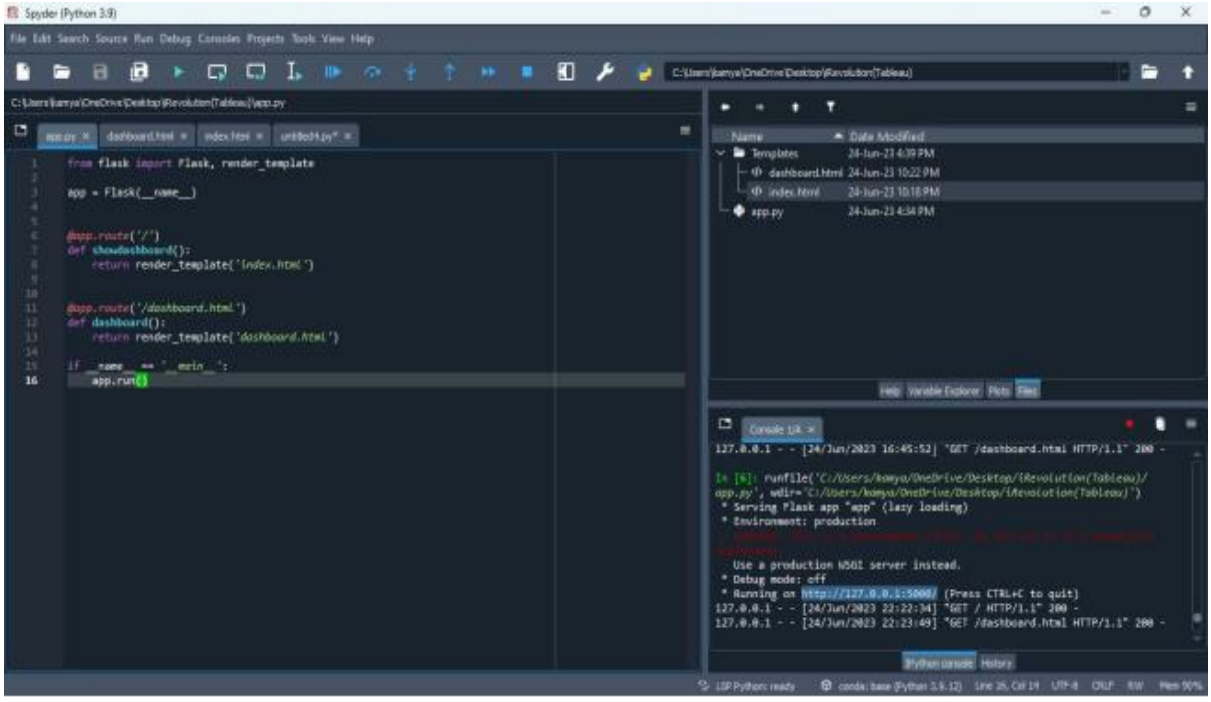


Tableau - Book1 - Tableau license expires in 14 days
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Data
Analytics
Tables
Pages
Filters
Measure Names
Marks
Measure Values
SUM(730)
Columns
Rows
Measure Values
Sheet 4
Value
Show Me
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
6 marks
1 row by 6 columns
SUM of Measure Values: 4.935


The New Chart Types pilot featuring Sankey and Radial charts is ending soon on June 30. If you participated in the pilot, tell us what you think! Share your feedback
iRevolution: Apple's iPhone Impact in India by Kamya Paliwal
More than 1 billion consumers currently use iPhones.
Since its initial launch, more than 1.9 billion iPhones have been sold.
iPhone sales in 2021 surpassed the 2015 peak, but declined in 2022 to 232.2 million units.
iPhone sales generated $205 billion revenue for Apple in 2022, more than Microsoft's total revenue. It sold 232 million iPhone units that year.
Comparative analysis amongst various other leading brands in the smartphone industry shows that iPhone is yet to make it's impact in India. Its 3% market share in the global market is depicted in the line-bar graph.
2%
2%
3%
3%
4%
Samsung Galaxy A03 Core
Samsung Galaxy A14 5G
Apple iPhone 13
vivo Y02
OnePlus Nord CE 2 Lite 5G
Country
India
Models
Apple iPhone 13
vivo Y02



Flask Integration



Dashboard, Report and Story embed with UI With Flask

Steps to Integrate BootstrapMade Template + Tableau in Flask

- Download Template
  - Go to BootstrapMade Free Templates.
  - Download a free template (e.g., *Arsha*).
  - Extract the .zip file.
- Organize Flask Project
  - Create folder tableau_flask/.
  - Inside it, create:
  - tableau_flask/

  ??? app.py

  ??? templates/

  ??? static/

- Copy files:

    - index.html ? into templates/
    - assets/, css/, js/, img/ ? into static/
- Fix File Paths
    - Open index.html.
    - Wherever you find assets/, replace it with static/.
    - Example:

  <link href="assets/css/style.css" rel="stylesheet">

- becomes

  <link href="static/assets/css/style.css" rel="stylesheet">

- Do the same for CSS, JS, and images.
- Embed Tableau Dashboard
    - In index.html, locate a main section (like <section id="hero">).
- Replace/add:

Project Demonstration & Documentation

Project Demonstration & Documentation involves presenting the project's functionality, features, and outcomes while providing clear written records, diagrams, and explanations to ensure understanding, usability, and reproducibility for stakeholders and future reference.

Record explanation Video for project end to end solution

Record a 5–7 minute screen-share video explaining the project end-to-end. Introduce your/team name, state the problem statement, highlight preprocessing steps, demonstrate Tableau dashboards, present the story, and showcase key insights clearly.

Project Documentation-Step by step project development procedure

Students have to write the project work step by step – starting from the problem, collecting data, cleaning it, making visuals/dashboards, finding results, and writing everything clearly in the document. Create document as per the template provided.

GROUP PHOTO

DEMO LINK :-
https://drive.google.com/file/d/1brsZsQ55NJIDNJscX3NA8rAm_GYAfAnY/view?usp=drivesdk